Multi-lingual and Multi-modal Retrieval for Audiovisual Media

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1. Introduction

We are now in an age when many resources are available for enjoying movies, such as Netflix and HBO Max. However, their search systems may not always provide optimal ranked results. For example, users whose native language is Chinese and who want to search for Western movies may not obtain the best results when their queries are written in Chinese. This represents a cross-lingual information retrieval problem in which the query and the relevant audiovisual content may be represented in different languages. In addition, users often want to enjoy a movie without searching for a specific title. Instead, they may have an exploratory information need, such as finding romantic movies, action movies, or movies made by well-known directors. Moreover, when users have a screenshot from a movie or an animation and want to identify it, popular platforms may not support this type of multimodal retrieval. Thus, to address these multilingual and multimodal retrieval issues, we want to create a platform that provides highly relevant search results and a better search experience while serving as an intermediate destination from which users can navigate to the source of the audiovisual content, regardless of whether they submit a textual or visual query.

2. IR Task Definition

The tasks for this project is multilingual and multimodal information retrieval. The system accepts either (1) a text query written in one of several supported languages or (2) a visual query, such as a poster or screenshots. The model shall return a ranked list of relevant films, television series, and animated work from a fixed document collection.

- Input: a natural-language query written in supported languages, including genre, name of the media, tags, setting, visual style, or mood. Also, it can be a visual query like poster or screenshot. Which means such input query parts can be toggled between text or graph.

- Document collection: one document per movie, television tile or animation work. The indexed core document contains English title, English overview for the document, genres, cast names, release year, media type (visual style), and original-language metadata.

- Output: a ranked list of media items, where each item mapped to a stable TMDB identifier and displayed with title, year, type, overview, and poster thumbnail if possible.

- Relevance: for known-item queries, the item the writer had in mind is relevant. For exploratory queries that can have multiple answers, relevance is graded as 0 to 5 (where 0 is the least relevant and 5 is the strongest relevant).

3. Data

The primary source will be The Movie Database (TMDB) API, which provides programmatic access to movie, television series, person, image, language, country, and translation data. After the registration for an API key, the API limit would be considered and the small scale of data would be used for dry run and evaluation. The metadata, images, or any related information would only retrieved from TMDB's current terms and would not link to unauthorized platforms or websites.

Collection procedure:

a. Retrieve some identifiers of well-known movies and television series from TMDB and check the limit of request from an free API key.

b. Fetch English details and available translations for each item

c. Retain items with a nonempty English title and overview and with sufficient translated metadata for query construction in selected languages
d. Deduplicate documents records by stable identifier and remove some genres which are not allowed to be shown in the course projects
e. Construct one English-language retrieval document per item
f. Store provenance, retrieval data, filtering reason, and TMDB identifier for reproducibility.
g. Download small poster images for those items for image queries.

Data and Queries	Planned Target
Selected items before filtering	50,000 to 100,000
Final English Documents	30,000 to 50,000
Movies	Around 40 to 50%
Television Series	Around 20 to 30%
Animation Work	Around 20 to 30%
Items with a usable poster	At least 80%
Chinese Queries	At least 100 known-answer queries + 30 exploratory
Spanish Queries	At least 100 known-answer queries + 30 exploratory
English Queries	At least 100 known-answer queries + 30 exploratory
Graded pooled judgements	Roughly 500 total

Query type	Query language and query itself	English gloss	Expected relevant document
Known-item Query	一部關於小女孩進入神靈世界 且父母變成豬的動畫電影	An animated movie where a girl enters a spirit world and her parents become pigs	Sprited Away (animation, Japanese origin)
Known-item Query	una serie de atracos con máscaras rojas y un profesor que dirige el plan	A heist series with red masks and a professor directing the plan	Money Heist (television, crime/drama)
Exploratory Query	Movies directed by Chritopher Nolan		Inception, Tenet
Exploratory Query	Funny family movie containing cooking or food		Chef, Ratatouille

4. Related Work

1. Tip of the Tongue Known-Item Retrieval: A Case Study in Movie Identification

How did they approach it? What makes their problem different from ours? Why do we think our approach will be better?
They analyzed 1,000 real movie-identification questions to understand how people describe forgotten movies, and tested traditional retrieval methods using Wikipedia plot descriptions, achieving only 13% accuracy in the top 10 results. The difference between us is that they are focusing on analyzing forgotten-movie searches using traditional retrieval methods, without considering multilingual queries, but our project could use cross-lingual semantic retrieval to help users find movies and animations through vague descriptions in different languages, potentially improving search accuracy.

2. XOR QA: Cross-lingual Open-Retrieval Question Answering

How did they approach it? What makes their problem different from ours? Why do we think our approach will be better?
They introduced a cross-lingual question-answering benchmark with around 40,000 questions across seven non-English languages, and tested machine translation and cross-lingual pretrained models to retrieve information across languages. The difference between their work and ours is that they focused on retrieving information to answer questions, rather than finding entertainment titles through incomplete descriptions. Our project could apply cross-lingual retrieval to help users find movies and animations regardless of the source language, while displaying the results in English.

3. Miracl: A multilingual retrieval dataset covering 18 diverse languages

How did they approach it? What makes their problem different from ours? Why do we think our approach will be better?
They developed a multilingual retrieval benchmark called MIRACL, covering 18 languages with around 78,000 queries and 726,000 relevance judgments to evaluate search performance across languages. The difference between their work and ours is that they focused on retrieving Wikipedia passages within the same language, while our project focuses on finding movies and animations across different languages. Our project could explore how language differences, vague descriptions, and translated movie titles affect search accuracy. However, our project would focus more specifically rather than building a large-scale benchmark like MIRACL since we have a limit of time

4. Colbert: Efficient and effective passage search via contextualized late interaction over bert

How did they approach it? What makes their problem different from ours? Why do we think our approach will be better?
They developed ColBERT, a BERT-based retrieval model that encodes queries and documents separately and uses late interaction to compare their similarity, making semantic search faster and more efficient. The difference between their work and ours is that they focused on improving general passage retrieval, while our project focuses on finding movies and animations using vague descriptions across different languages. Our project could explore whether semantic retrieval performs better than traditional methods like BM25 for forgotten-movie searches, while also considering multilingual queries.

5. Learning transferable visual models from natural language supervision

How did they approach it? What makes their problem different from ours? Why do we think our approach will be better?
They developed CLIP, a model that connects images with natural language descriptions to support image-text matching and recognition without additional task-specific training. The difference between their work and ours is that they focused on general image-text matching, while our project focuses on finding movies and animations using descriptions, images, and multilingual queries. Our project could use CLIP-like models to help users search for movies through screenshots or visual details, combining them with movie metadata to potentially improve search accuracy.

5. Evaluation and Results

We will evaluate our search system by testing around 500+ queries in English+Chinese, or more languages like Japanese, Spanish, including movie titles and vague descriptions, and possibly compare several search methods like: TF-IDF, BM25, PivotedNormalization, or probably our own designed ones. For each query, we will manually rate how relevant the search results are on a scale of 1 to 5, and will mainly use nDCG@10 to measure how well the system ranks relevant movies, and MRR to measure how quickly it finds the correct movie. We will also compare results across different languages to see whether multilingual semantic search performs better than traditional keyword search.

6. Work Plan

Our team has defined the project's definition, primary research objectives, potential datasets, and explored relevant literature on multilingual and semantic information retrieval. We have also outlined our proposed evaluation methods.

From October 12 to November 9, we will prioritize dataset collection, preprocessing, construction of a multilingual query evaluation set, and implementation of random and BM25 retrieval baselines. By the November 9 Project Progress Check, we aim to have a cleaned dataset, a functioning baseline retrieval system, and preliminary quantitative evaluation results; From November 9 to December 6, we will implement multilingual semantic retrieval and, if feasible, hybrid retrieval. We will evaluate performance using Recall@10 and Mean Reciprocal Rank (MRR), comparing retrieval effectiveness across query languages and incomplete descriptions; From December 7 to December 21, we will conduct error analysis, finalize our experiments, document findings, and prepare the final report and recorded demonstration.

Responsibilities will be divided between dataset preparation and baseline development on one side, and multilingual semantic retrieval and query evaluation on the other. Both team members will jointly review experimental results and contribute to the final report.

To ensure feasibility, we will prioritize text-based retrieval and quantitative evaluation. Image-based retrieval, geographic availability filtering, and advanced interface features will remain optional extensions, dependent on progress with the core system.

Week	Dates	Objectives and Measurable Deliverables
1	Oct 12–18	Finalize dataset selection and permissions. Collect 500–1,000 titles and assess metadata and language coverage.
2	Oct 19–25	Clean metadata, remove duplicates, map multilingual titles, and expand to 5,000–10,000 titles if feasible.
3	Oct 26–Nov 1	Implement Random and BM25 baselines. Prepare at least 50 evaluation queries.
4	Nov 2–8	Evaluate baselines using Recall@10 and MRR. Document preliminary results anderrors.
Milestone	Nov 9	Project Progress Check
5	Nov 9–15	Implement multilingual semantic retrieval and compare against BM25.
6	Nov 16–22	Expand to 100–150 evaluation queries in English, Chinese, and Japanese.

7 8 9	Nov 23–29 Nov 30–Dec 6 Dec 7–13	Implement hybrid retrieval if feasible and compare retrieval approaches. Evaluate language-specific performance and produce 3–5 tables or figures. Analyze retrieval errors, finalize experiments, and develop a simple demonstration.
10	Dec 14–20	Complete the report, documentation, and video demonstration.
Deadline	Dec 21	Submit the Final Project Report and Video Recording.

7. Chat History

prompting and chat history